

Exploratory Testing Report

1. Application Under Test

Application: EchoGPT – Multi-AI Chat Sidebar Chrome Extension and Android Application

Testing Type: Exploratory Testing

Testing Duration: Approximately 30–45 minutes per exploratory session

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2. Testing Objective

The objective of exploratory testing was to evaluate the overall usability, functionality, state management, error handling, and user experience of the EchoGPT Chrome Extension and Android application beyond predefined functional test cases.

The testing focused on discovering unexpected behavior, state persistence issues, validation problems, error-handling weaknesses, and usability improvements.

3. Test Environment

Web Extension

- **Operating System:** Windows 11
- **Browser:** Google Chrome
- **Application:** EchoGPT Chrome Extension
- **Account:** Free account

Android Application

- **Platform:** Android
- **Application:** EchoChat / EchoGPT Android application
- **Account:** Free account

4. Exploratory Testing Approach

The following exploratory techniques were used:

- Feature exploration
- Positive and negative testing
- Boundary and edge-case testing
- State transition testing
- Input validation testing
- Navigation testing
- Error-message evaluation

- Session/state persistence testing
- UI/UX observation
- Cross-platform observation

The tester interacted with different features without relying exclusively on predefined test cases and followed unexpected behavior when observed.

5. Areas Explored

The following areas were explored during testing:

Web Extension

- Chat and New Chat
- Model selection and switching
- Chat history
- File attachment
- Translate
- Read
- Read Aloud
- Write
- Compare
- MCP / Connectors
- Settings
- Theme and extension controls

Android Application

- New Chat
- History
- Previous conversation reopening
- Subscription
- Image Studio
- Compare
- Connectors
- Support
- Discord navigation
- Model selection
- Chat input
- Keyboard behavior
- Screen rotation
- Background/resume behavior

6. Key Observations

Several functional and state-management issues were identified during exploratory testing.

6.1 State Persistence

Some feature states were retained after starting a New Chat.

Observed examples included:

- Previously attached files remaining after New Chat.
- Unsent chat drafts remaining after New Chat.
- Previous Translate results remaining after New Chat.
- Previous Read results remaining after New Chat.
- Unsent Write content remaining after New Chat.

These behaviors can make it difficult for users to determine whether they are working in a completely new session.

6.2 Translate

The Translate feature successfully produced translation results and the Copy function worked correctly.

However:

- The language swap control did not exchange the source and target languages.
- Previous translation results could remain after clearing the source input.
- Previous translation results could remain after starting a New Chat.

6.3 Read Feature

The Read feature successfully processed supported text files and displayed results.

An unsupported image file produced an appropriate file-type validation message.

However, the Read Aloud control did not produce audible playback during testing.

6.4 Compare

The Compare interface allowed model selection, but after selecting a model, the expected comparison response was not generated during testing.

6.5 MCP / Connectors

Connector validation was explored using valid and invalid URL formats.

The application correctly displayed a validation message for an invalid URL format:

“URL must start with https://”

For a syntactically valid but unreachable URL, the application displayed:

“Could not reach the MCP server: fetch failed”

The messages provide basic error information, although additional recovery guidance could improve the experience.

6.6 Settings

Settings generally opened correctly and displayed account information.

An intermittent authentication/navigation issue was previously observed where Settings redirected to the Sign In screen instead of opening Settings. The behavior was not consistently reproducible.

6.7 Android Application

Several Android functions were explored, including:

- Chat history
- Conversation reopening
- Model switching
- Subscription duration selection
- Image Studio
- Compare
- Connectors
- Support
- Discord navigation
- Share functionality
- Keyboard and back behavior
- Multiline input
- Screen rotation
- Background/resume behavior

The tested navigation and input-related functions generally behaved as expected.

7. Edge Cases Explored

The following edge cases were specifically considered:

1. Starting a New Chat while an unsent message exists.
2. Starting a New Chat while a file is attached.
3. Clearing Translate input after generating a result.
4. Reopening Translate after starting a New Chat.
5. Reopening Read after processing a file.
6. Entering an invalid connector URL.
7. Attempting to connect to an unreachable MCP server.
8. Uploading an unsupported file type.
9. Using multiline chat input.
10. Switching between AI models.
11. Navigating away from and returning to feature screens.
12. Testing Android behavior after screen rotation and background/resume.

8. Bugs Discovered

The exploratory testing activities contributed to identifying and confirming the following issues:

- Translate language swap control does not work.
- Read Aloud produces no audio.
- Settings intermittently redirects to Sign In.
- Compare with selected model does not generate a response.
- New Chat does not clear previously attached file.
- New Chat retains unsent chat draft.
- Previous Translate result remains after clearing source input.
- New Chat retains previous Translate result.
- New Chat retains previous Read result.
- New Chat retains unsent Write input.

These issues are documented separately in the Bug Reports section of the assignment.

9. UI/UX Observations

The exploratory sessions also identified several opportunities for improving usability:

- Consistent branding between Web and Android.
- Better visual feedback for Translate language swapping.
- Clear playback state for Read Aloud.
- Loading/processing feedback for Compare.
- Clear visual confirmation when starting a New Chat.
- More visible supported file-type information.
- Clear file-size/upload constraints.
- Proactive guidance for connector URL formatting.
- Dedicated Clear/Reset controls for feature results.
- Retry/troubleshooting guidance for MCP connection failures.

These recommendations are documented separately in the UI/UX Review section.

10. Recommendations

Based on the exploratory testing, the following improvements are recommended:

1. Implement a reliable state-reset mechanism when New Chat is selected.
2. Provide clear visual feedback for asynchronous operations such as Compare and Read Aloud.
3. Add dedicated Clear/Reset controls to feature-based workflows.
4. Improve error recovery by providing actionable next steps.
5. Display file-upload requirements before users select files.
6. Maintain consistent branding across Web and Android applications.
7. Review authentication/session handling around Settings.
8. Ensure feature results do not unintentionally persist across unrelated sessions.

11. Conclusion

The exploratory testing covered major conversational, utility, navigation, input, file-processing, state-management, and error-handling areas of the EchoGPT Web Extension and Android application.

The testing identified multiple genuine functional issues as well as UI/UX improvement opportunities. Particular attention was required for state persistence between New Chat sessions, feature-result handling, error recovery, and feedback for asynchronous operations.

The findings from exploratory testing were used to supplement the formal functional test cases, bug reports, and UI/UX review included in this assignment.